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Particulate matter deposition on roadside plants and the importance of leaf trait combinations

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1. Abstract & Scope

Vehicular traffic is a primary source of urban particulate matter (PM), which accumulates on roadside plants. We evaluated the relative influence of species-specific leaf traits and trait combinations on PM deposition on 16 native plant species in Sydney, Australia. Leaves were Agitated in water to measure SPM. Heavy metals (Cu, Cr, Mn, Al, Fe) were analyzed using ICP-AES. Results show significant differences in PM deposition among species. Leaf shape was the strongest predictor of deposition, with lanceolate-leaved species accumulating more PM than other shapes. While foliar trichomes (hairs) generally increased PM capture, their effect was sometimes overridden by leaf cuticular waxes, which create a hydrophobic barrier preventing initial wet deposition.

2. Methodology & Experimental Parameters

Roadside sampling within 2m of arterial lanes in Sydney. 60-second reverse-osmosis water wash-off. Cascade vacuum filtration. Trace metal extraction via nitric/hydrochloric acid digestion and ICP-AES analysis.

3. Core Findings & Data Synthesis

Westringia fruticosa (pubescent lanceolate shrub) accumulated the highest PM loads. PERSOONIA and DODONAEA (long-petioled, large leaves) had high wind flutter and low retention. Acacia parramattensis accumulated significant Chromium (0.51 mg/kg).

4. Strategic Relevance to JU and Kolkata Planning

Highlights the importance of matching leaf traits (stiff lanceolate leaves with low flutter) to vehicular roadside dust characteristics.

